

Theoretical Foundations for 3D Reconstruction of Sports Games with Limited Visual Data Settings

The 2022 FIFA World Cup was the first major football tournaments to introduce a semi-automated offside technology (SAOT) to reconstruct the 3D scene of the players and the ball on a football pitch when an offside call is made. Despite being a major breakthrough in football refereeing and broadcasting, this technology remains inaccessible to many less developed countries and regional tournaments. This ISC project aims to recreate such a technology in the context of low camera coverage and low quality data. To achieve that, this project looks into the theoretical foundations into building a minimum-viable-product (MVP) styled three-step pipeline that is able to reconstruct the full football field model, from segmentation of field lines, reconstructing the graph skeleton representing the field, to recovering the homography transformation from a template field to the target field.